EMS Readiness Optimization for Manhattan

Executive Presentation

Slide 1: Title

Optimizing Ambulance Staging in Manhattan

A Simulation-Based Approach to Reducing EMS Response Times

- EMS Optimization Research Team
- March 2026
- Prepared for FDNY Operations Leadership

Slide 2: The Problem

Current EMS Response Performance is Unacceptable

The current uniform allocation policy (P0):
- Distributes ambulances **equally** across 48 firehouses
- **Ignores** where crashes actually happen
- Results in **8.08-minute average response time**
- Only **64.4%** of calls get response within 8 minutes

Why this matters:
- Manhattan has ~3.48 MVC calls per hour (30,500+ per year)
- Midtown generates 3–5× more calls than Upper Manhattan
- Every minute of delay increases mortality risk for critical patients

> "The difference between a 2-minute and 8-minute response can be the difference between life and death."

Slide 3: Current State — P0 Performance

Uniform Allocation: Equal ≠ Equitable

Metric	P0 Value	Target
Mean Response Time	**8.08 min**	< 4 min
P90 Response Time	**19.47 min**	< 8 min
8-min Coverage	**64.4%**	> 95%
Utilization	9.1%	—

Root Cause: Spatial mismatch between where ambulances are staged and where crashes occur.

- Midtown precincts (highest demand) are underserved
- Upper Manhattan precincts (lowest demand) have excess capacity
- The system is not capacity-constrained—it's location-constrained

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Slide 4: Our Solution — Optimized Allocation (P2)

Smart Staging: Put Ambulances Where They're Needed

Approach:
1. Analyzed **2.24 million** historical crash records
2. Built a demand model capturing hourly and geographic patterns
3. Used **mathematical optimization** to find the best allocation
4. Validated with **1,440 simulation runs**

P2 Allocation Strategy:
- Concentrates units in **high-demand areas** (Midtown, Lower Manhattan)
- Maintains **minimum coverage** in low-demand areas
- Uses the **same 20 ambulances** — just staged differently
- Implementable as a **simple shift-change staging plan**

Slide 5: Key Results — Head-to-Head Comparison

P2 Delivers Transformative Performance Improvements

Metric	P0 (Current)	P2 (Optimized)	Improvement
Mean Response Time	8.08 min	**2.57 min**	**↓ 68%**
P90 Response Time	19.47 min	**3.76 min**	**↓ 81%**
8-min Coverage	64.4%	**99.6%**	**↑ 35 pp**
Fleet Used	20 units	20 units	**Same fleet**

Statistical confidence:
- All improvements significant at $p < 0.001$
- Effect sizes are "Large" (Cohen's $d > 28$)
- Based on 30 independent simulation replications
- 95% CI for mean RT improvement: [5.41, 5.61] minutes

Slide 6: Performance Improvements Visualized

Response Time Distribution Shift

P0 Distribution: Wide spread, 8.08 min mean, long tail to 21+ min
P2 Distribution: Tight cluster around 2.57 min, 99.6% under 8 min

Key Visual Insights (see results/figures/):

- `pub_fig1_policy_comparison.png` — Side-by-side policy comparison
- `pub_fig2_fleet_sensitivity.png` — How fleet size affects each policy
- `pub_fig5_performance_heatmap.png` — Performance across all scenarios

The bottom line: P2 doesn't just improve the average—it virtually eliminates long waits.

Slide 7: Robustness Analysis

P2 Dominates Under ALL Tested Conditions

Fleet Size Sensitivity (K = 15 to 40):
- P2 achieves >99% coverage with just K=25 units
- P0 requires K=40 to reach 99% coverage
- **P2 with 15 units outperforms P0 with 40 units**

Demand Variation (0.5× to 2.0×):
- P2 mean RT ranges 2.44–2.85 min (stable)
- P0 mean RT ranges 7.80–8.58 min (degrades)
- Policy rankings unchanged across all demand levels

Service Time Sensitivity (20–30 min mean):
- No impact on policy rankings
- Response time unaffected by service time variations
- Results hold under all operational assumptions tested

Slide 8: Implementation Roadmap

Three-Phase Deployment Plan

Phase 1 — Pilot (Months 1–3)
- Deploy P2 at **5 highest-impact firehouses** (Midtown)
- Monitor mean RT, 8-min coverage, and utilization daily
- Success criterion: ≥15% improvement in pilot area response times

Phase 2 — Expansion (Months 3–6)
- Expand to **15–20 firehouses** based on pilot results
- Calibrate model with real dispatch data
- Begin road-network travel time integration

Phase 3 — Full Deployment (Months 6–12)
- Roll out to **all 48 Manhattan firehouses**
- Implement shift-specific allocations
- Establish continuous optimization cycle

Requirements: No new equipment, no new technology. Just repositioning existing ambulances.

Slide 9: Expected Benefits

Annual Impact Estimate (K=20 units)

Benefit	Value
Calls per year affected	**~30,500** MVC calls
Mean RT reduction per call	**5.5 minutes**
Total annual time saved	**167,750 minutes** (2,796 hours)
Additional calls within 8-min target	**~10,700** per year
Effective fleet multiplier	**3×** (20 units → equivalent of 60)

Beyond the Numbers:

- **Lives saved** through faster defibrillation and trauma response
- **Reduced disability** from faster stroke and cardiac care
- **Improved equity** — better coverage for underserved areas
- **Cost avoidance** — same performance with fewer units needed

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Slide 10: Recommendations and Next Steps

Immediate Recommendations

1. ✅ **Approve P2 adoption** as the standard ambulance staging policy for Manhattan
2. ✅ **Launch pilot** at Midtown firehouses within 30 days
3. ✅ **Establish monitoring** dashboard with real-time KPI tracking
4. ✅ **Brief dispatch leadership** on new staging locations

Future Opportunities

- **Dynamic repositioning**: Adjust staging in real-time based on current unit positions
- **Multi-borough expansion**: Extend optimization to Brooklyn, Queens, Bronx, Staten Island
- **All-incident integration**: Include medical emergencies, fires, and hazmat calls
- **CAD integration**: Embed optimization recommendations in dispatch software

Questions?

Contact: EMS Optimization Research Team
Repository: github.com/cnsp/ems-optimization
Full Technical Report: docs/technical_report.md

This analysis is based on 1,440 simulation experiments, 2.24 million historical crash records, and rigorous statistical validation. All results are reproducible from the project repository.